INFO8006 Introduction to Artificial Intelligence

Exercise session 4: Probabilistic reasoning

Recap

Bayesian networks

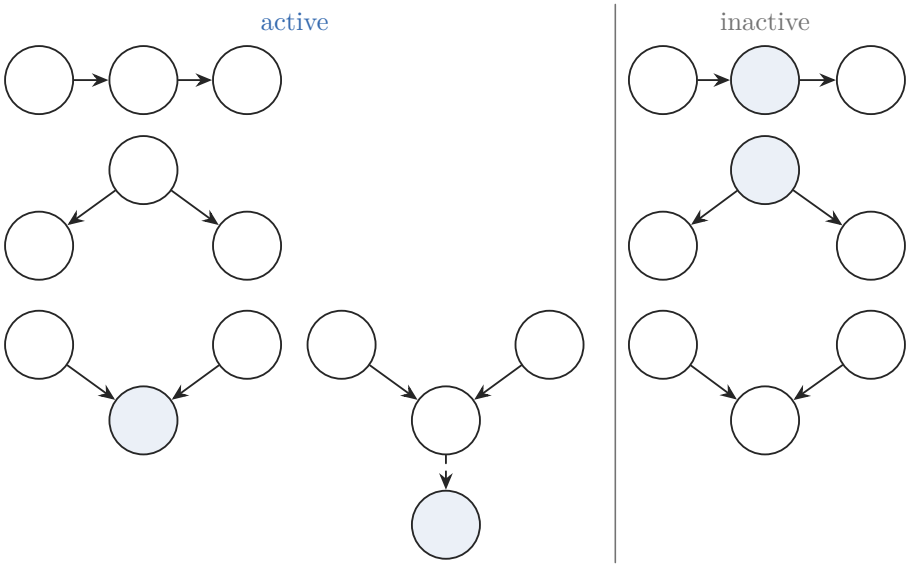
The state s and the percepts e of an agent are assignments of random variables $X_1, \ldots, X_n$, with joint distribution $P(s, e) = P(s)P(e \mid s)$. A *Bayesian network* is a directed acyclic graph with one node per variable X_i, annotated with a *conditional probability table* (CPT) $\mathbf{P}(X_i \mid \text{parents}(X_i))$. It asserts the *local Markov property* $X_i \perp \text{non-descendants}(X_i) \mid \text{parents}(X_i)$, hence

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid \text{parents}(X_i)).$$

d-separation

$X \perp Y \mid \mathbf{Z}$ is guaranteed if every undirected path between X and Y is *inactive*; a path is *active* iff all its triples are:

Triple		Active iff
Cascade	$A \to B \to C$ or $A \leftarrow B \leftarrow C$	B unobserved
Common parent	$A \leftarrow B \to C$	B unobserved
v-structure	$A \to B \leftarrow C$	B or a descendant of B observed



Given its *Markov blanket* (parents, children, other parents of its children), a variable is independent of all others.

Inference (optional)

For a *query* Q, *evidence* $\mathbf{E} = \mathbf{e}$ and *hidden* variables $\mathbf{H}$,

$$\mathbf{P}(Q \mid \mathbf{e}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathbf{P}(Q, \mathbf{h}, \mathbf{e}), \qquad Z = \sum_{q} \sum_{\mathbf{h}} P(q, \mathbf{h}, \mathbf{e}).$$

- *Enumeration* evaluates the sums depth-first: space $O(n)$, time $O(d^n)$ for n variables of d values.
- *Variable elimination joins factors*, initially the CPTs given $\mathbf{e}$, by pointwise products, and *eliminates* hidden variables by summing over their values. Its cost, that of the largest factor, depends on the ordering; the best one is NP-hard to find.

Parameter learning

For a parameterized family $\mathbf{P}(X \mid \theta)$ and i.i.d. data $\mathbf{d} = \{x_1, \ldots, x_N\}$:

Likelihood	$P(\mathbf{d} \mid \theta) = \prod_{j=1}^{N} P(x_j \mid \theta), \quad L(\theta) = \log P(\mathbf{d} \mid \theta)$
Maximum likelihood (MLE)	$\theta^*_{\text{MLE}} = \arg\max_\theta L(\theta)$, often from $\partial L / \partial \theta = 0$
Posterior	$\mathbf{P}(\theta \mid \mathbf{d}) \propto \mathbf{P}(\mathbf{d} \mid \theta)\mathbf{P}(\theta)$, for a prior $\mathbf{P}(\theta)$
Maximum a posteriori (MAP)	$\theta^*_{\text{MAP}} = \arg\max_\theta P(\mathbf{d} \mid \theta)P(\theta)$

Exercises

In session: exercises 1, 2.

Exercise 1 Nuclear power plant (AIMA ch. 13)

A monitoring program supervises the core of a nuclear power station. An alarm sounds when a temperature gauge, which measures the temperature of the core, exceeds a given threshold, and the program reasons about the actual temperature from the gauge and the alarm, either of which may be faulty. Consider the Boolean variables A (alarm sounds), F_A (alarm is faulty), and F_G (gauge is faulty) and the multi-valued nodes G (gauge reading) and T (actual core temperature).

(a) Draw a Bayesian network for this domain, given that the gauge is more likely to fail when the core temperature gets too high.

(b) From your network topology, can you conclude $T \perp F_A \mid F_G$ and $T \perp F_A \mid F_G, A$?

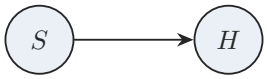
(c) Suppose there are just two possible actual and measured temperatures: low (l) and high (h). The probability that the gauge gives the correct temperature is x when it is working, but y when it is faulty. Give the conditional probability table associated with G.

(d) Suppose the alarm is always triggered by high measured temperatures, unless it is faulty, in which case it never sounds. Give the conditional probability table associated with A.

(e) Suppose the gauge is not faulty and the alarm is triggered. Calculate an expression for the probability that the temperature of the core is too high, in terms of the various conditional probabilities in the network.

Exercise 2 Energy of an inference server

A scheduling program manages a data center in which a GPU server answers the requests sent to a language model. To plan the power supply, the scheduler models the energy the server consumes as a function of its load, with the following network.



where S is a random variable denoting the number of requests the server receives in an hour, and H a random variable denoting the energy, in kilowatt-hours, that it consumes during that hour. The scheduler chooses the following relations for each variable:

$$P(S = s) = \texttt{Poisson}(s \mid \lambda) = \frac{\lambda^s e^{-\lambda}}{s!} \tag{1}$$

$$H = \omega S + H_0 + \mathcal{N}(0, \sigma^2) \tag{2}$$

where H_0 is the energy the server consumes when idle, which is known, and ω is the energy consumed per request. The term $\mathcal{N}(0, \sigma^2)$ is a Gaussian noise of mean 0 and variance σ^2, where the Gaussian density of mean μ and variance σ^2 is

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

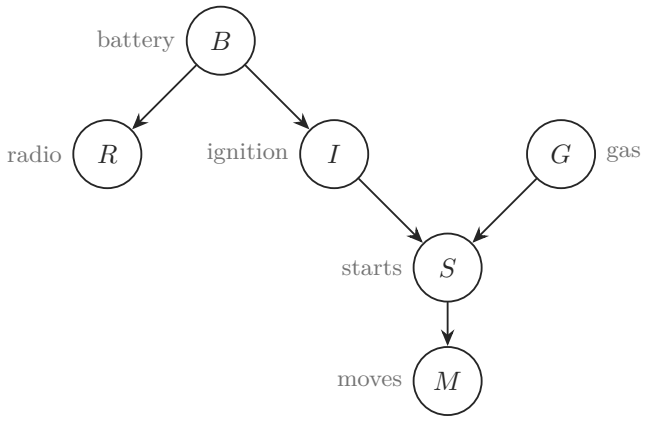
(a) Identify the parameters of the Bayesian network and give the joint log-likelihood of one pair (s_i, h_i) given the model.

(b) The scheduler collects a set $\mathbf{d} = \{(s_i, h_i)\}_{i=1}^{N}$ of independent observations. Derive the maximum likelihood estimate of the parameters.

(c) The scheduler has just started to collect observations, and maximum likelihood can overfit when they are few. It therefore treats the parameter λ, about which it has no strong knowledge, as uncertain. Using a prior $\lambda \sim \texttt{Gamma}(\alpha, \beta)$, what would be the posterior distribution $p(\lambda \mid \mathbf{s})$, with $\mathbf{s} = \{s_i\}_{i=1}^{N}$?
Hint: $\texttt{Gamma}(x \mid \alpha, \beta) = \frac{1}{Z} x^{\alpha-1} e^{-\beta x}$, with Z a normalizing constant.

(d) Knowing that the mean, the variance and the mode of $\texttt{Gamma}(\alpha, \beta)$ are respectively $\frac{\alpha}{\beta}$, $\frac{\alpha}{\beta^2}$ and $\frac{\alpha - 1}{\beta}$, interpret how they evolve w.r.t. N between the prior and the posterior.

Exercise 3 Car diagnosis (AIMA ch. 13)
Consider the following Bayesian network, which describes some features of a car's electrical system and engine. Each variable is Boolean, and the true value indicates that the corresponding aspect of the vehicle is in working order.



(a) Extend the network with the Boolean variables *IcyWeather* and *StarterMotor*.

(b) According to your knowledge of cars, give reasonable conditional probability tables for all the nodes.

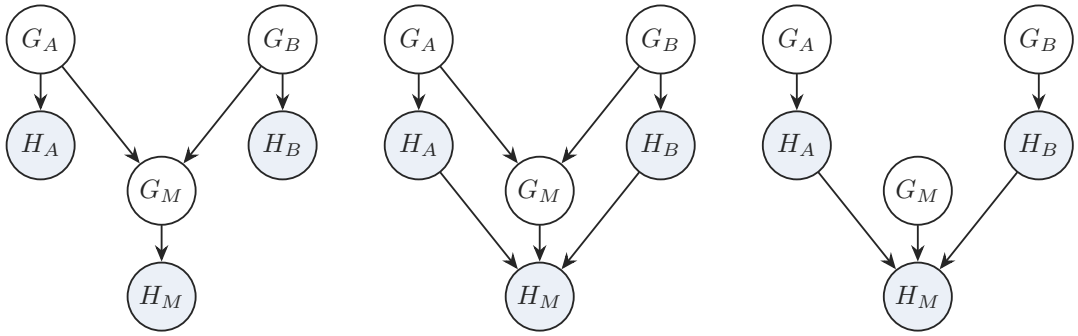
(c) How many independent values are contained in the joint probability distribution for eight Boolean nodes, assuming that no conditional independence relations are known to hold among them?

(d) How many independent probability values do your network tables contain?

Exercise 4 Evolutionary model merging (AIMA ch. 13)
Evolutionary model merging builds a new language model M from two parent models A and B: each component of M is copied from one of the parents. Consider one component, and let G_x be the language it was mostly trained on in model x, with possible values e (English) or f (French). A common *hypothesis* is that the default language of a model follows from this component by a simple mechanism: the language H_x in which model x answers by default is the same as G_x with some probability s.

Furthermore, M copies the component from either parent with equal probability, and a random perturbation flips it with a small nonzero probability m.

(a) Which of the following networks claim that $\mathbf{P}(G_M \mid G_A, G_B) = \mathbf{P}(G_M)$?



(b) Which of the networks make independence claims that are consistent with the hypothesis about the default language?

(c) Write down the CPT for the G_M node in the first network, in terms of s and m.

(d) Suppose that $P(G_A = f) = P(G_B = f) = q$. In the first network, derive an expression for $P(G_M = f)$ in terms of s, m and q.

(e) Over many generations of merges, we expect the distribution of the component to be the same in the parents and in the merged models. Use this to calculate the value of q. Only about 10 % of the merged models answer in French by default: explain why the hypothesis described at the beginning of this question must be wrong.

Exercise 5 Lidar calibration

A self-driving car calibrates its new lidar against a reference target before driving. The distance μ_y at which the lidar sees the target is uncertain: the specification of the calibration station gives the prior $\mu_y \sim \mathcal{N}(\mu_l, \sigma_l^2)$. Each reading y_i of the lidar is Gaussian, $y_i \sim \mathcal{N}(\mu_y, \sigma^2)$, with a noise variance σ^2 known from its datasheet.

The calibration program of the car records a calibration set $\mathcal{Y}_c$ of C readings, and wants to predict the next reading y of the lidar. We study how much $\mathcal{Y}_c$ reduces the uncertainty about this reading. Before the calibration, the reading is predicted by the *prior predictive* distribution $p(y) = \int p(y \mid \mu_y) p(\mu_y)\, d\mu_y$; after it, by the *posterior predictive* distribution $p(y \mid \mathcal{Y}_c) = \int p(y \mid \mu_y) p(\mu_y \mid \mathcal{Y}_c)\, d\mu_y$.

You may use the following identities. If $p(x) = \mathcal{N}(x \mid m, S)$ and $p(z \mid x) = \mathcal{N}(z \mid hx + u, R)$, then x and z are jointly Gaussian,

$$p(x, z) = \mathcal{N}\left(\begin{pmatrix} x \\ z \end{pmatrix} \middle| \begin{pmatrix} m \\ hm + u \end{pmatrix}, \begin{pmatrix} S & hS \\ hS & h^2 S + R \end{pmatrix} \right).$$

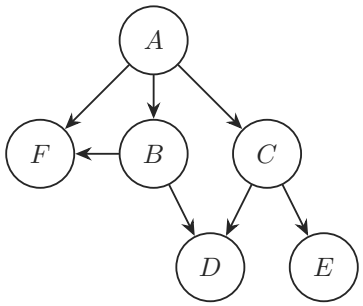
If x and z are jointly Gaussian with means a and b, variances A and B, and covariance K, then

$$p(z) = \mathcal{N}(z \mid b, B), \qquad p(x \mid z) = \mathcal{N}\left(x \middle| a + \frac{K}{B}(z - b), A - \frac{K^2}{B} \right).$$

(a) Compute the prior and posterior predictive distributions $p(y)$ and $p(y \mid \mathcal{Y}_c)$ if the calibration set holds one reading, *i.e.* $\mathcal{Y}_c = \{y_o\}$.

(b) Looking at the predictive uncertainty, what is the benefit of incorporating $\mathcal{Y}_c$?

Exercise 6 Independence and variable elimination
Which of the following statements does the structure of the network below guarantee?



(a) $\mathbf{P}(B, C) = \mathbf{P}(B)\mathbf{P}(C)$

(b) $\mathbf{P}(B, C \mid A) = \mathbf{P}(B \mid A)\mathbf{P}(C \mid A)$

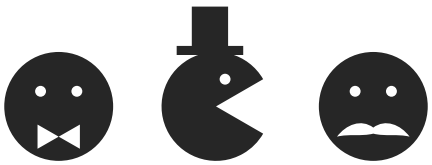
(c) $\mathbf{P}(F, E) = \mathbf{P}(F)\mathbf{P}(E)$

(d) $\mathbf{P}(F \mid A, D, E) = \mathbf{P}(F \mid A, D)$

(e) $\mathbf{P}(B, E) = \sum_{a,c,d,f} P(a)\mathbf{P}(B \mid a)P(c \mid a)\mathbf{P}(d \mid B, c)\mathbf{P}(E \mid c)\mathbf{P}(f \mid a, B)$

(f) Use inference by variable elimination to compute $\mathbf{P}(E \mid A = 1, B = 1)$.

Exercise 7 Recognizing Pacman (CS188, Spring 2014)
The ghosts of the maze are controlled by a program that watches the maze through cameras and must recognize Pacman in their pictures. It learns to do so from a set of pictures of Pacman, and of ghosts as negative examples.



Because the pictures are black and white, and were taken from various angles, the program identifies Pacman from salient features: the presence of a bowtie B, hat H or mustache M. The following table summarizes the content of the pictures. Each feature takes values in $\{0, 1\}$, where 0 and 1 mean the feature is respectively absent and present. The subject of the picture is described by a random variable $S \in \{0, 1\}$, where 0 is a ghost and 1 is Pacman.

B	H	M	S
0	0	0	1
1	0	0	0
1	0	1	1
1	1	0	0
0	1	0	0
1	1	1	1

(a) Suppose the program is a naive Bayes classifier. Draw the Bayesian network that represents the dependencies between S, B, H and M for the program.

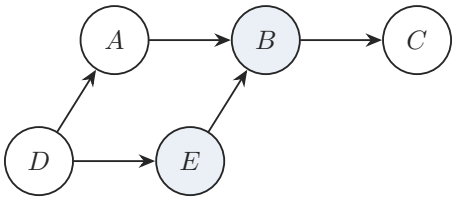
(b) Write the Bayesian classification rule for this problem, *i.e.* the formula that given a data point (b, h, m) returns the most likely subject. Write the formula in terms of conditional and prior probabilities. What does the formula become under the assumptions of the program?

(c) What are the parameters of this model? Give estimates of these parameters according to the pictures of the table.

(d) Pacman eventually shows up in front of a camera wearing a bowtie, but no hat or mustache. Will the program recognize him?

Quiz

1. Consider the Bayesian network shown below. We want to infer $\mathbf{P}(A \mid b, e)$ where A is the query variable, B and E are evidence variables, and C and D are hidden variables. Which of the following statements is true?



   - ◯ $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A)P(b \mid e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$
   - ◯ $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A, e)\mathbf{P}(A \mid d)P(e \mid d)$
   - ◯ $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A, e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$
   - ◯ $\mathbf{P}(A \mid b, e) \propto \sum_b \sum_e P(c \mid b)\mathbf{P}(A \mid b, e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$

2. In a Bayesian network
   - ◯ independence between two variables is guaranteed if all paths connecting them are inactive.
   - ◯ dependence between two variables can be guaranteed using d-separation.
   - ◯ a path between two nodes is active if it contains at least one active triple.
   - ◯ a cascade triple is active if the center node is observed.

3. Consider a Bayesian network with a single binary variable, following a Bernoulli distribution. If we observe T positive realizations and F negative ones,
   - ◯ the maximum likelihood estimate of the positive probability is $\frac{T}{F}$.
   - ◯ the maximum likelihood estimate of the positive probability is $\frac{T+F}{T}$.
   - ◯ the maximum likelihood estimate of the positive probability is $\frac{T}{T+F}$.
   - ◯ none of the above.